

Wave 1 Directional Retraining Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- Campaign Name: `wave1_directional_retraining_campaign_2026_05_06_16_07_16`
- Closeout Timestamp: `2026-05-07-13-48-50`
- Campaign Output Directory: `output/training_campaigns/wave1/directional_retraining/2026-05-06-16-58-54_wave1_directional_retraining_campaign_2026_05_06_16_07_16`
- Completed Runs: `15`
- Failed Runs: `0`
- All `15` planned Wave 1 runs are now closed across `global`, `forward`, and `backward` variants.
- The protected `doc/running/active_training_campaign.yaml` file was intentionally left untouched because it still tracks a separate `RCIM Model-Bank Reproduction` campaign.

Closeout Actions

- Repaired directional metadata in `0` tree metrics snapshots so registry-facing artifacts now preserve `base_model_family`, `training_variant`, and direction flags consistently.
- Rebuilt the affected family registries, the campaign leaderboard, the campaign best-run snapshots, and the program best registry from the repaired metrics.
- Archived one ONNX export plus one Python artifact for every `wave 1` family/scope winner under `models/exported/`, together with scope-local inventories, dataset provenance, and source-run snapshots.
- Refreshed the canonical `wave 1` closeout report and regenerated the training-results master summary from the updated registries.

Campaign Ranking

Rank	Family	Scope	Run	Model Type	Val MAE [deg]	Test MAE [deg]	Test RMSE [deg]
1	tree	forward	te_hist_gbr_tabular_Fw	hist_gradient_boosting	0.002666	0.002845	0.003476
2	tree	global	te_hist_gbr_tabular_global	hist_gradient_boosting	0.002719	0.002885	0.003607
3	tree	backward	te_hist_gbr_tabular_Bw	hist_gradient_boosting	0.002698	0.003087	0.003850
4	harmonic_regression	forward	te_harmonic_order12_linear	harmonic_regression	0.002811	0.003129	0.003567

Rank	Family	Scope	Run	Model Type	Val MAE [deg]	Test MAE [deg]	Test RMSE [deg]
			conditioned_ recovery_ Fw				
5	feedforward	global	te_ feedforward_ stride1_ high_ compute_ long_ remote_ global	feedforw ard	0.003056	0.003150	0.003603
6	feedforward	backward	te_ feedforward_ stride1_ high_ compute_ long_ remote_ Bw	feedforw ard	0.003049	0.003262	0.003749
7	periodic_mlp	global	te_ periodic_ mlp_h04_ standard_ global	periodic _mlp	0.002985	0.003349	0.003916
8	residual_harmonic_mlp	global	te_ residual_ h12_deep_ joint_ wave1_ global	residual _harmonic _mlp	0.003115	0.003420	0.003931
9	periodic_mlp	forward	te_ periodic_ mlp_h04_ standard_ Fw	periodic _mlp	0.002848	0.003432	0.004023
10	residual_harmonic_mlp	backward	te_ residual_ h12_deep_ joint_ wave1_Bw	residual _harmonic _mlp	0.003110	0.003493	0.004108
11	harmonic_regression	backward	te_ harmonic_ order12_ linear_ conditioned_on recovery_ Bw	harmonic _regressi	0.003701	0.003524	0.004080
12	periodic_mlp	backward	te_ periodic_ mlp_h04_ standard_ Bw	periodic _mlp	0.003154	0.003525	0.004132
13	residual_harmonic_mlp	forward	te_ residual_ h12_deep_ joint_ wave1_Fw	residual _harmonic _mlp	0.002852	0.003530	0.004145
14	feedforward	forward	te_ feedforward_ stride1_ high_ compute_ long_ remote_ Fw	feedforw ard	0.002915	0.003563	0.004009

Rank	Family	Scope	Run	Model Type	Val MAE [deg]	Test MAE [deg]	Test RMSE [deg]
15	harmonic_regression	global	te_harmonic_order12_linear_conditioned_on_recovery_global	harmonic_regression	0.017017	0.020779	0.022403

Family Directional Surface

- Current campaign-best entry: `te_hist_gbr_tabular_fw` from family `tree_fw` with `test_mae = 0.002845 deg`.

Family	Global Test MAE [deg]	Forward Test MAE [deg]	Backward Test MAE [deg]	Best Scope
tree	0.002885	0.002845	0.003087	forward
residual_harmonic_mlp	0.003152	0.003530	0.003493	global
feedforward	0.003150	0.003563	0.003262	global
periodic_mlp	0.003317	0.003432	0.003525	global
harmonic_regression	0.020779	0.003129	0.003524	forward

Exported Model Archive

- Export root: `models/exported`
- Root inventory: `models/exported/wave1_directional_retraining_export_inventory.yaml`
- Each family now exposes `global/`, `forward/`, and `backward/` subfolders, each containing `python/`, `onnx/`, `reference_inventory.yaml`, `dataset_snapshot_manifest.yaml`, and `source_runs/<run_instance_id>/` provenance snapshots.

Family	Scope	Python Artifact	ONNX Artifact	Provenance Bundle
tree	global	<code>models/exported/tree/global/python/tree_model.pkl</code>	<code>models/exported/tree/global/onnx/model.onnx</code>	<code>models/exported/tree/global/reference_inventory.yaml</code>
tree	forward	<code>models/exported/tree/forward/python/tree_model.pkl</code>	<code>models/exported/tree/forward/onnx/model.onnx</code>	<code>models/exported/tree/forward/reference_inventory.yaml</code>
tree	backward	<code>models/exported/tree/backward/python/tree_model.pkl</code>	<code>models/exported/tree/backward/onnx/model.onnx</code>	<code>models/exported/tree/backward/reference_inventory.yaml</code>
residual_harmonic_mlp	global	<code>models/exported/residual_harmonic_mlp/global/python/residual_harmonic_mlp-epoch=077-val_mae=0.00302384.ckpt</code>	<code>models/exported/residual_harmonic_mlp/global/onnx/model.onnx</code>	<code>models/exported/residual_harmonic_mlp/global/reference_inventory.yaml</code>
residual_harmonic_mlp	forward	<code>models/exported/residual_harmonic_mlp/forward/python/residual_harmonic_mlp-epoch=018-val_mae=0.00285191.ckpt</code>	<code>models/exported/residual_harmonic_mlp/forward/onnx/model.onnx</code>	<code>models/exported/residual_harmonic_mlp/forward/reference_inventory.yaml</code>

Family	Scope	Python Artifact	ONNX Artifact	Provenance Bundle
residual_harmonic_mlp	backward	models/exported/residual_harmonic_mlp/backward/python/residual_harmonic_mlp-epoch=037-val_mae=0.00310962.ckpt	models/exported/residual_harmonic_mlp/backward/onnx/model.onnx	models/exported/residual_harmonic_mlp/backward/reference_inventory.yaml
feedforward	global	models/exported/feedforward/global/python/feedforward-epoch=180-val_mae=0.00305586.ckpt	models/exported/feedforward/global/onnx/model.onnx	models/exported/feedforward/global/reference_inventory.yaml
feedforward	forward	models/exported/feedforward/forward/python/feedforward-epoch=033-val_mae=0.00291539.ckpt	models/exported/feedforward/forward/onnx/model.onnx	models/exported/feedforward/forward/reference_inventory.yaml
feedforward	backward	models/exported/feedforward/backward/python/feedforward-epoch=093-val_mae=0.00304864.ckpt	models/exported/feedforward/backward/onnx/model.onnx	models/exported/feedforward/backward/reference_inventory.yaml
periodic_mlp	global	models/exported/periodic_mlp/global/python/periodic_mlp-epoch=031-val_mae=0.00309735.ckpt	models/exported/periodic_mlp/global/onnx/model.onnx	models/exported/periodic_mlp/global/reference_inventory.yaml
periodic_mlp	forward	models/exported/periodic_mlp/forward/python/periodic_mlp-epoch=022-val_mae=0.00284801.ckpt	models/exported/periodic_mlp/forward/onnx/model.onnx	models/exported/periodic_mlp/forward/reference_inventory.yaml
periodic_mlp	backward	models/exported/periodic_mlp/backward/python/periodic_mlp-epoch=049-val_mae=0.00315372.ckpt	models/exported/periodic_mlp/backward/onnx/model.onnx	models/exported/periodic_mlp/backward/reference_inventory.yaml
harmonic_regression	global	models/exported/harmonic_regression/global/python/harmonic_regression-epoch=018-val_mae=0.01701703.ckpt	models/exported/harmonic_regression/global/onnx/model.onnx	models/exported/harmonic_regression/global/reference_inventory.yaml
harmonic_regression	forward	models/exported/harmonic_regression/forward/python/harmonic_regression-epoch=068-val_mae=0.00281060.ckpt	models/exported/harmonic_regression/forward/onnx/model.onnx	models/exported/harmonic_regression/forward/reference_inventory.yaml
harmonic_regression	backward	models/exported/harmonic_regression/backward/python/harmonic_regression-epoch=019-val_mae=0.00370070.ckpt	models/exported/harmonic_regression/backward/onnx/model.onnx	models/exported/harmonic_regression/backward/reference_inventory.yaml

Canonical Follow-Through

- Wave 1 summary refreshed: doc/reports/analysis/wave1/Wave 1 - Closeout Status.md
- master summary refreshed: doc/reports/analysis/Training Results Master Summary.md
- campaign leaderboard refreshed: output/training_campaigns/wave1/directional_retraining/2026-05-06-16-58-54_wave1_directional_retraining_campaign_2026_05_06_16_07_16/campaign_leaderboard.yaml
- campaign best run refreshed: output/training_campaigns/wave1/directional_retraining/2026-05-06-16-58-54_wave1_directional_retraining_campaign_2026_05_06_16_07_16/campaign_best_run.yaml
- program best registry: output/registries/program/current_best_solution.yaml